

2016.0 RANGE ROVER (LG), 100-00

GENERAL INFORMATION

DESCRIPTION AND OPERATION

TOUCH SCREEN (TS)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable.
Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be logged in the Touch Screen (TS). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual.

For additional information, refer to: [Information and Entertainment System](#)

(415-00 Information and Entertainment System - General Information, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1087-83	LIN Bus "A" - Value of signal protection calculation incorrect	Rear view camera LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear view camera LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B1087-86	LIN Bus "A" - Signal invalid	Rear view camera LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear view camera LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B1087-87	LIN Bus "A" - Missing message	Rear view camera LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear view camera LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B1087-88	LIN Bus "A" - Bus off	Rear view camera LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	Refer to the electrical circuit diagrams and check the rear view camera LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B137F-31	Steering Wheel Left Switch Pack - No signal	Steering wheel switchpack left power or ground circuit open circuit, high resistance	- Refer to the electrical circuit diagrams and test the steering wheel switchpack left power and ground circuits for open circuit, high resistance - Refer to the electrical circuit

		▪ LIN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Steering wheel switchpack left failure	diagrams and check the LIN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
B1D21-13	Remote Control Switch - Circuit open	▪ Steering wheel switchpack circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the steering wheel switchpack circuit for open circuit, high resistance (rotate steering wheel during tests)
U0001-87	High Speed CAN Communication Bus - Missing message	▪ Private high speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and test the telephone interface module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the private high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0010-00	Medium Speed CAN Communication Bus - No sub type information	▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: If CAN network integrity test fails, more information can be found here: 4: Electrical, 418: Electrical Distribution, 418-00 Module Communications Network (Diagnosis and Testing, Communications network) ▪ Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to

			ground, short circuit to power, open circuit, high resistance
U0010-48	Medium Speed CAN Communication Bus - Supervision software failure	Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: If CAN network integrity test fails, more information can be found here: 4: Electrical, 418: Electrical Distribution, 418-00 Module Communications Network (Diagnosis and Testing, Communications network) Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0010-87	Medium Speed CAN Communication Bus - Missing message	- Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Body control module/gateway module assembly power or ground circuit open circuit, high resistance - Parking aid control module power or ground circuit open circuit, high resistance - Driver seat module power or ground circuit	- Using the manufacturer approved diagnostic system, complete a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the body control module/gateway module assembly power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the parking aid control module power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the driver seat module power and ground circuits for open circuit, high resistance - Refer to the electrical circuit diagrams and check the passenger

		open circuit, high resistance ▪ Passenger seat module power or ground circuit open circuit, high resistance ▪ Rear junction box power or ground circuit open circuit, high resistance	seat module power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the rear junction box power and ground circuits for open circuit, high resistance
U0074-00	Control Module Communication Bus "B" Off - No sub type information	▪ Telephone interface module power or ground circuit open circuit, high resistance ▪ Navigation control module power or ground circuit open circuit, high resistance ▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance	▪ Refer to the electrical circuit diagrams and test the telephone interface module power and ground circuits for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and test the navigation control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance
U0142-00	Lost Communication With Body Control Module "B" - No sub type information	▪ Rear view camera power or ground circuit open circuit, high resistance ▪ Private high speed CAN bus circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Rear view	▪ Refer to the electrical circuit diagrams and test the rear view camera power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the private high speed CAN bus circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the body

		camera	control module for related DTCs and refer to the relevant DTC index
U0159-00	Lost Communication With Parking Assist Control Module "A" - No sub type information	▪ Parking aid control module power or ground circuit open circuit, high resistance ▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Parking aid system fault	▪ Refer to the electrical circuit diagrams and check the parking aid control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the parking aid control module for related DTCs and refer to the relevant DTC index
U0164-00	Lost Communication With HVAC Control Module - No sub type information	▪ Automatic temperature control module power or ground circuit open circuit, high resistance ▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Automatic temperature control system fault	▪ Refer to the electrical circuit diagrams and check the automatic temperature control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the automatic temperature control module for related DTCs and refer to the relevant DTC index
U0166-00	Lost Communication With Auxiliary Heater Control Module - No sub type	▪ Fuel fired booster heater power or ground circuit open circuit, high resistance	▪ Refer to the electrical circuit diagrams and check the fuel fired booster heater power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved

	information	▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Fuel fired booster heater system fault	diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the fuel fired booster heater for related DTCs and refer to the relevant DTC index
U0184-00	Lost Communication With Radio - No sub type information	▪ Integrated audio module power or ground circuit open circuit, high resistance ▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Integrated audio module system fault	▪ Refer to the electrical circuit diagrams and check the integrated audio module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the integrated audio module for related DTCs and refer to the relevant DTC index
U0184-4A	Lost Communication With Radio - Incorrect component installed	▪ The serial number of the integrated audio module does not match the serial number stored in the touch screen	▪ Check the serial number of the integrated audio module is valid (as installed at the factory), if the part has been installed by a dealer the installation routine has not been performed correctly. Configure the integrated audio module using the manufacturer approved diagnostic system. Clear the DTC and retest the system
U0186-00	Lost Communication With Audio Amplifier "A" - No sub type information	▪ Audio amplifier module power or ground circuit open circuit, high resistance	

		▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Audio amplifier module system fault	NOTE: This DTC may be stored even though no fault condition is present and should be ignored unless the customer has reported an information and entertainment system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis ▪ Refer to the electrical circuit diagrams and check the audio amplifier module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the audio amplifier module for related DTCs and refer to the relevant DTC index
U0186-4A	Lost Communication With Audio Amplifier "A" - Incorrect component installed	▪ The serial number of the audio amplifier module does not match the serial number stored in the touch screen	▪ Check the serial number of the audio amplifier module is valid (as installed at the factory), if the part has been installed by a dealer the installation routine has not been performed correctly. Configure the audio amplifier module using the manufacturer approved diagnostic system. Clear the DTC and retest the system
U0191-00	Lost Communication With Television - No sub type information	▪ TV control module power or ground circuit open circuit, high resistance ▪ Medium speed	

		CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance TV control module system fault	NOTE: This DTC may be stored even though no fault condition is present and should be ignored unless the customer has reported an information and entertainment system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis - Refer to the electrical circuit diagrams and check the TV control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the TV control module for related DTCs and refer to the relevant DTC index
U0191-4A	Lost Communication With Television - Incorrect component installed	The serial number of the TV control module does not match the serial number stored in the touch screen	Check the serial number of the TV control module is valid (as installed at the factory), if the part has been installed by a dealer the installation routine has not been performed correctly. Configure the TV control module using the manufacturer approved diagnostic system. Clear the DTC and retest the system
U0193-00	Lost Communication With "Digital Audio Control Module A" - No sub type information	- Satellite radio control module power or ground circuit open circuit, high resistance - Medium speed CAN bus	

		(comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Satellite radio system fault	NOTE: This DTC may be stored even though no fault condition is present and should be ignored unless the customer has reported an information and entertainment system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis ▪ Refer to the electrical circuit diagrams and check the satellite radio control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the satellite radio control module for related DTCs and refer to the relevant DTC index
U0193-4A	Lost Communication With "Digital Audio Control Module A" - Incorrect component installed	▪ The serial number of the satellite radio control module does not match the serial number stored in the touch screen	▪ Check the serial number of the satellite radio control module is valid (as installed at the factory), if the part has been installed by a dealer the installation routine has not been performed correctly. Configure the satellite radio control module using the manufacturer approved diagnostic system. Clear the DTC and retest the system
U0194-00	Lost Communication With "Digital Audio Control Module B" - No sub type	▪ Digital radio control module power or ground circuit open circuit, high resistance	

	information	▪ Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Digital radio system fault	NOTE: This DTC may be stored even though no fault condition is present and should be ignored unless the customer has reported an information and entertainment system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis ▪ Refer to the electrical circuit diagrams and check the digital radio control module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the digital radio control module for related DTCs and refer to the relevant DTC index
U0194-4A	Lost Communication With "Digital Audio Control Module B" - Incorrect component installed	▪ The serial number of the digital radio control module does not match the serial number stored in the touch screen	▪ Check the serial number of the digital radio control module is valid (as installed at the factory), if the part has been installed by a dealer the installation routine has not been performed correctly. Configure the digital radio control module using the manufacturer approved diagnostic system. Clear the DTC and retest the system
U0196-00	Lost Communication With Entertainment Control Module-Rear	▪ Rear seat entertainment control module power or ground circuit open circuit,	

	"A" - No sub type information	high resistance - Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Rear seat entertainment system fault	NOTE: This DTC may be stored even though no fault condition is present and should be ignored unless the customer has reported an information and entertainment system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis - Refer to the electrical circuit diagrams and check the rear seat entertainment control module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the rear seat entertainment control module for related DTCs and refer to the relevant DTC index
U0196-4A	Lost Communication With Entertainment Control Module-Rear "A" - Incorrect component installed	The serial number of the rear seat entertainment control module does not match the serial number stored in the touch screen	Check the serial number of the rear seat entertainment control module is valid (as installed at the factory), if the part has been installed by a dealer the installation routine has not been performed correctly. Configure the rear seat entertainment control module using the manufacturer approved diagnostic system. Clear the DTC and retest the system
U0208-00	Lost Communication With "Seat Control Module A" -	Driver seat module power or ground circuit open circuit, high resistance	Refer to the electrical circuit diagrams and test the driver seat module power and ground circuits for open circuit, high resistance

	No sub type information	▪ Medium speed CAN bus (body) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Driver seat module fault	▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the medium speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the driver seat module for related DTCs and refer to the relevant DTC index
U0209-00	Lost Communication With "Seat Control Module B" - No sub type information	▪ Passenger seat module power or ground circuit open circuit, high resistance ▪ Medium speed CAN bus (body) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Passenger seat module fault	▪ Refer to the electrical circuit diagrams and test the passenger seat module power and ground circuits for open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and test the medium speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Using the manufacturer approved diagnostic system, check the passenger seat module for related DTCs and refer to the relevant DTC index
U0232-00	Lost Communication With Side Obstacle Detection Control Module - Left - No sub type information	▪ Left blindspot monitoring control module power or ground circuit open circuit, high resistance ▪ Medium speed CAN bus (body) circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Left blindspot monitoring	NOTE: Check to ensure the vehicle is fitted with blind spot monitoring. If blind spot monitoring is not fitted, then due to hardware/software incompatibility this DTC may be stored even though no fault condition is present and should be ignored. ▪ Refer to the electrical circuit diagrams and check the left blindspot monitoring control

		system fault	module power and ground circuits for open circuit, high resistance ■ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the manufacturer approved diagnostic system, check the left blindspot monitoring control module for related DTCs and refer to the relevant DTC index
U0233-00	Lost Communication With Side Obstacle Detection Control Module - Right - No sub type information	■ Right blindspot monitoring control module power or ground circuit open circuit, high resistance ■ Medium speed CAN bus (body) circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Right blindspot monitoring system fault	NOTE: Check to ensure the vehicle is fitted with blind spot monitoring. If blind spot monitoring is not fitted, then due to hardware/software incompatibility this DTC may be stored even though no fault condition is present and should be ignored. ■ Refer to the electrical circuit diagrams and check the blindspot monitoring control module power and ground circuits for open circuit, high resistance ■ Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (body) circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Using the manufacturer approved diagnostic system, check the right blindspot monitoring control module for related DTCs and refer to the relevant DTC index

U023B-00	Lost Communication With Image Processing Module B - No sub type information	- General proximity sensor module power or ground circuit open circuit, high resistance - Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance - General proximity sensor system fault	- Refer to the electrical circuit diagrams and check the general proximity sensor module power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the general proximity sensor module control module for related DTCs and refer to the relevant DTC index
U025D-00	Lost Communication With Front Controls Interface Module "B" - No sub type information	- Integrated control panel power or ground circuit open circuit, high resistance - Medium speed CAN bus (comfort) circuit short circuit to ground, short circuit to power, open circuit, high resistance - Integrated control panel system fault	- Refer to the electrical circuit diagrams and check the integrated control panel power and ground circuits for open circuit, high resistance - Using the manufacturer approved diagnostic system, perform a CAN network integrity test. Refer to the electrical circuit diagrams and check the medium speed CAN bus (comfort) circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Using the manufacturer approved diagnostic system, check the integrated control panel for related DTCs and refer to the relevant DTC index
U0300-00	Internal Control Module Software Incompatibility - No sub type information	Touch screen is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the touch screen with the latest level software
U0300-51	Internal Control Module Software Incompatibility - Not	Touch screen is not configured correctly	Using the manufacturer approved diagnostic system, re-configure the touch screen with the latest level software

	programmed		
U1A24-87	MOST Ring Complete. No Communication - Missing message	Control module on MOST network not communicating	NOTE: This DTC may be stored even though no fault condition is present and should be ignored unless the customer has reported an information and entertainment system concern. Clear the DTC and retest. Verify the customer concern prior to diagnosis Refer to the relevant section of the workshop manual and test the MOST network
U2003-87	Fibre Optic Communication Bus - Missing message	MOST network failure	Refer to the relevant section of the workshop manual and test the MOST network
U3000-4B	Control Module - Over temperature	Touch screen high internal temperature detected	Allow the system to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new touch screen
U3000-98	Control Module - Component or system over temperature	Control module on MOST network high internal temperature detected	NOTE: This DTC is set when the touch screen receives a system shut down request from another control module on the MOST network Allow the system to cool. Using the manufacturer approved diagnostic system, clear the DTCs and retest. If the fault persists, install a new control module as necessary
U3003-62	Battery Voltage - Signal	Mismatch between the	Refer to the electrical circuit diagrams and check the touch

compare failure

voltage at the
touch screen
and the voltage
value broadcast
on the CAN bus

screen power and ground circuits for
open circuit, high resistance